

# Locally Polynomial Regression

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## 1. Locally polynomial regression

## **Locally polynomial regression**

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# Locally polynomial estimator

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Locally  $k$ th-order polynomial regression approximates  $m(u)$  as:

$$m(u) \approx m(x) + m'(x)(u - x) + \cdots + \frac{1}{k!}m^{(k)}(x)(u - x)^k$$

for  $u \approx x$

$$\operatorname{argmin}_{\tilde{m}(x) \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (Y_i - \tilde{m}(X_i))^2 K_h(X_i - x)$$

Locally polynomial estimators:

$$\begin{pmatrix} \hat{m}(x) \\ \hat{m}^{(1)}(x) \\ \vdots \\ \hat{m}^{(k)}(x) \end{pmatrix} = \operatorname{argmin}_{a \in \mathbb{R}^{k+1}} \frac{1}{n} \sum_{i=1}^n \left( Y_i - \sum_{j=0}^k \frac{a_j}{j!} (X_i - x)^j \right)^2 K_h(X_i - x)$$

# Explicit Form

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Let  $D_x \in \mathbb{R}^{n \times (k+1)}$ ,  $W_x = \text{diag}(K_h(X_i - x))$ :

$$D_x^\top = \begin{pmatrix} 1 & \dots & 1 \\ \frac{X_1 - x}{1!h} & \dots & \frac{X_n - x}{1!h} \\ \vdots & \ddots & \vdots \\ \frac{(X_1 - x)^k}{k!h^k} & \dots & \frac{(X_n - x)^k}{k!h^k} \end{pmatrix}$$

Then

$$\begin{aligned} \hat{\mathbf{m}}(x) &= \begin{pmatrix} \hat{m}(x) \\ h\hat{m}'(x) \\ \vdots \\ h^k \hat{m}^{(k)}(x) \end{pmatrix} = \underset{a \in \mathbb{R}^{k+1}}{\text{argmin}} (Y - D_x a)^\top W_x (Y - D_x a) \\ &= (D_x^\top W_x D_x)^{-1} D_x^\top W_x Y \end{aligned}$$

# Non-Singularity of $D_x^\top W_x D_x$

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- The estimator exists only if  $D_x^\top W_x D_x$  is non-singular

$$D_x^\top W_x D_x = \left( \frac{\sum_{r=1}^n (X_r - x)^{i+j} K_h(X_r - x)}{i!j!h^{i+j}} \right)_{ij}$$

$$\lambda^\top D_x^\top W_x D_x \lambda = \sum_{i=1}^n K_h(X_i - x) \left[ \lambda_0 + \lambda_1 \left( \frac{X_i - x}{h} \right) + \cdots + \frac{\lambda_k}{k!} \left( \frac{X_i - x}{h} \right)^k \right]^2 = 0$$

implies  $\lambda_0 = \cdots = \lambda_k = 0$  if  $K_h(X_i - x) > 0$  for  $(k+1)$  distinct  $X_i$

# Minimal Bandwidth

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Minimal bandwidth defined by:

$$h_{\min} = \max(X_{(k+1)}, 1 - X_{(n-k)}, \frac{1}{2}(X_{(k+2)} - X_{(1)}), \dots,$$

$$\dots \frac{1}{2}(X_{(n-1)} - X_{(n-k-2)}), \frac{1}{2}(X_{(n)} - X_{(n-k-1)}))$$

$h > h_{\min}$  makes at least  $(k+1)$  distinct  $X_i$ s are in  $[x-h, x+h]$  for every  $x \in [0, 1]$ .

# Existence of Local Polynomial Estimators

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If  $h \rightarrow 0$  and  $nh \rightarrow \infty$  as  $n \rightarrow \infty$  and  $0 < p_{\min, \epsilon}(x) < p_{\max, \epsilon}(x) < \infty$  for a small  $\epsilon > 0$ , then  $\mathbb{P}(D_x^\top W_x D_x \text{ is invertible}) \rightarrow 1$ .

$\mathbb{P}(\text{There are at most } k \text{ distinct } X_i \in [x - h, x + h])$

$$\begin{aligned} &= \sum_{j=0}^k \binom{n}{j} q_x^j (1 - q_x)^{n-j} \\ &= (1 - q_x)^{n-k} \left[ \binom{n}{0} (1 - q_x) + \binom{n}{1} q_x (1 - q_x)^{k-1} + \dots + \binom{n}{k} q_x^k \right] \\ &\leq \exp(-nh \cdot p_{\min, \epsilon}(x)) \cdot (k + 1) \cdot (2nh \cdot p_{\max, \epsilon}(x))^k \text{ for all large } n \\ &\rightarrow 0 \end{aligned}$$

using  $\binom{n}{i} \leq n^i$ ,  $(1 - x)^n \leq \exp(-nx)$  for  $x \geq 0$

# Pointwise Error: Conditional Bias

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Assume that  $m$  is  $(k+1)$ -times continuously differentiable at  $x$  and that  $p$  is continuous at  $x$  and  $p(x) > 0$ . Suppose that  $h \rightarrow 0$  and  $nh \rightarrow \infty$  as  $n \rightarrow \infty$ . Then,

$$E(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) - \mathbf{m}(x) = h^{k+1} \cdot \frac{1}{(k+1)!} \cdot \mathbf{M}(x)^{-1} \boldsymbol{\mu}(x) \cdot m^{(k+1)}(x) + o_p(h^{k+1}).$$

where

$$\mathbf{M}(x) = \begin{pmatrix} \frac{1}{0!0!} \mu_0(x) & \frac{1}{0!1!} \mu_1(x) & \cdots & \frac{1}{0!k!} \mu_k(x) \\ \frac{1}{1!0!} \mu_1(x) & \frac{1}{1!1!} \mu_2(x) & \cdots & \frac{1}{1!k!} \mu_{k+1}(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{k!0!} \mu_k(x) & \frac{1}{k!1!} \mu_{k+1}(x) & \cdots & \frac{1}{k!k!} \mu_{2k}(x) \end{pmatrix}, \boldsymbol{\mu}(x) = \begin{pmatrix} \frac{1}{0!} \mu_{k+1}(x) \\ \frac{1}{1!} \mu_{k+2}(x) \\ \vdots \\ \frac{1}{k!} \mu_{2k+1}(x) \end{pmatrix}$$

$$E(\hat{\mathbf{m}}(x) \mid X_n) - \mathbf{m}(x) = \left( D_x^\top W_x D_x \right)^{-1} D_x^\top W_x (\mathbf{m} - D_x \mathbf{m}(x))$$

$$\frac{D_x^\top W_x D_x}{n} = \frac{1}{n} \left( \frac{\sum_{r=1}^n (X_r - x)^{i+j} K_h(X_r - x)}{i!j!h^{i+j}} \right)_{ij} = \left( \frac{\hat{\nu}_{i+j}(x)}{i!j!} \right)_{ij}, i, j = 0, \dots, k$$

$$\begin{aligned} \frac{D_x^\top W_x (\mathbf{m} - D_x \mathbf{m}(x))}{n} &= \frac{1}{n} \left( \sum_{r=1}^n \frac{(X_r - x)^i}{i!h^i} K_h(X_r - x) \left\{ m(X_r) - \sum_{j=0}^k \frac{(X_r - x)^j m^{(j)}(x)}{j!} \right\} \right)_i \\ &= \left( \frac{\hat{\beta}_i(x)}{i!} \right)_i, i = 0, \dots, k \end{aligned}$$

where  $\hat{\nu}_\ell(x) := n^{-1} \sum_{i=1}^n \left( \frac{X_i - x}{h} \right)^\ell K_h(X_i - x), \ell = 0, \dots, 2k$

$\hat{\beta}_\ell(x) := n^{-1} \sum_{i=1}^n \left( \frac{X_i - x}{h} \right)^\ell K_h(X_i - x) \left( m(X_i) - \sum_{j=0}^k \frac{(X_i - x)^j m^{(j)}(x)}{j!} \right), \ell = 0, \dots, k$

$$\begin{aligned}
 E\hat{\nu}_\ell(x) &= E \left[ \left( \frac{X_1 - x}{h} \right)^\ell K_h(X_1 - x) \right] = \frac{1}{h^\ell} K_h^- * \left( (\cdot - x)^\ell p(\cdot) \right) (x) \\
 &= \frac{\mu_\ell(x)}{\ell!} \frac{\ell!}{h^\ell} p(x) h^\ell + o(1) = \mu_\ell(x) p(x) + o(1) \\
 \text{Var} \hat{\nu}_\ell(x) &= \frac{1}{n} \text{Var} \left[ \left( \frac{X_1 - x}{h} \right)^\ell K_h(X_1 - x) \right] \\
 &= \frac{1}{nh^{2\ell+1}} (K^2)_h^- * \left( (\cdot - x)^{2\ell} p(\cdot) \right) (x) - \frac{1}{n} E \left[ \left( \frac{X_1 - x}{h} \right)^\ell K_h(X_1 - x) \right]^2 \\
 &= \frac{1}{nh} \left( \frac{\mu_{2\ell}(x)}{2\ell!} \frac{1}{h^{2\ell}} p(x) h^{2\ell} + o(h^{2\ell}) \right) + o\left(\frac{1}{nh}\right) = \frac{1}{nh} \mu_{2\ell}(x) p(x) + o\left(\frac{1}{nh}\right) \\
 \hat{\nu}_\ell(x) &= \mu_\ell(x) p(x) + o_p(1)
 \end{aligned}$$

$$\frac{1}{n}D_x^T W_x D_x = \left( \frac{\hat{\nu}_{i+j}(x)}{i!j!} \right)_{ij} = p(x) \left( \frac{\mu_{i+j}(x)}{i!j!} \right)_{ij} + o_p(1) = p(x)\mathbf{M}(x) + o_p(1)$$

$$\begin{aligned}
 E\hat{\beta}_\ell(x) &= E \left[ \left( \frac{X_1 - x}{h} \right)^\ell K_h(X_1 - x) \left( m(X_1) - \sum_{j=0}^k \frac{(X_1 - x)^j m^{(j)}(x)}{j!} \right) \right] \\
 &= \frac{1}{h^\ell} K_h^- * \left( (\cdot - x)^\ell \left( m(\cdot) - \sum_{j=0}^k \frac{(\cdot - x)^j m^{(j)}(x)}{j!} \right) p(\cdot) \right) (x) \\
 &= \frac{\mu_{\ell+k+1}(x)}{(\ell+k+1)!} \binom{\ell+k+1}{\ell, (k+1)} \frac{\ell!}{h^\ell} m^{(k+1)}(x) p(x) h^{\ell+k+1} + o(h^{k+1}) = \mu_{\ell+k+1}(x) p(x) + o(h^{k+1}) \\
 &= \frac{\mu_{\ell+k+1}(x)}{(k+1)!} m^{(k+1)}(x) p(x) h^{k+1} + o(h^{k+1})
 \end{aligned}$$

$$\begin{aligned}
 Var \hat{\beta}_\ell(x) &= \frac{1}{n} Var \left[ \left( \frac{X_1 - x}{h} \right)^\ell K_h(X_1 - x) \left( m(X_1) - \sum_{j=0}^k \frac{(X_1 - x)^j m^{(j)}(x)}{j!} \right) \right] \\
 &= \frac{1}{nh^{2\ell+1}} (K^2)_h^- * \left( (\cdot - x)^{2\ell} \left( m(\cdot) - \sum_{j=0}^k \frac{(\cdot - x)^j m^{(j)}(x)}{j!} \right)^2 p(\cdot) \right) (x) \\
 &\quad - \frac{1}{n} E \left[ \left( \frac{X_1 - x}{h} \right)^\ell K_h(X_1 - x) \right]^2 \\
 &= \frac{1}{nh} \frac{\kappa_{2\ell+2k+2}(x)}{(2\ell + 2k + 2)!} \binom{2\ell + 2k + 2}{2\ell, 2(k+1)} \frac{(2\ell)!}{h^{2\ell}} \left( m^{(k+1)}(x) \right)^2 p(x) h^{2\ell+2k+2} + o(h^{2k+2}) \\
 &\quad O_p(\sqrt{Var \hat{\beta}_\ell(x)}) = O_p(n^{-1/2} h^{-1/2} h^{k+1}) = o_p(h^{k+1})
 \end{aligned}$$

$$\begin{aligned}
 \frac{1}{n} D_x^T W_x(\mathbf{m} - D_x \mathbf{m}(x)) &= \left( \frac{\hat{\beta}_i(x)}{i!} \right)_i \\
 &= \frac{1}{(k+1)!} \frac{\mu_{i+k+1}(x)}{i!} m^{(k+1)}(x) p(x) h^{k+1} + o_p(h^{k+1}) \\
 &= \frac{p(x)}{(k+1)!} h^{k+1} m^{(k+1)}(x) \boldsymbol{\mu}(x) + o_p(h^{k+1})
 \end{aligned}$$

$$\begin{aligned}
 E(\hat{\mathbf{m}}(x) \mid X_n) - \mathbf{m}(x) &= \left( n^{-1} D_x^\top W_x D_x \right)^{-1} n^{-1} D_x^\top W_x (\mathbf{m} - D_x \mathbf{m}(x)) \\
 &= [p(x) \mathbf{M}(x) + o_p(1)]^{-1} \left( \frac{p(x)}{(k+1)!} h^{k+1} m^{(k+1)}(x) \boldsymbol{\mu}(x) + o_p(h^{k+1}) \right) \\
 &= \left( \frac{1}{p(x)} [\mathbf{M}(x)]^{-1} + o_p(1) \right) \left( \frac{p(x)}{(k+1)!} h^{k+1} m^{(k+1)}(x) \boldsymbol{\mu}(x) + o_p(h^{k+1}) \right) \\
 &= h^{k+1} \cdot \frac{1}{(k+1)!} \cdot \mathbf{M}(x)^{-1} \boldsymbol{\mu}(x) \cdot m^{(k+1)}(x) + o_p(h^{k+1})
 \end{aligned}$$

## j-th derivative

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Recall

$$\mathbf{m}(x) = \begin{pmatrix} m(x) \\ hm'(x) \\ \vdots \\ h^k m^{(k)}(x) \end{pmatrix}$$

$$E(\hat{m}^{(j)}(x) \mid \mathbb{X}_n) - m^{(j)}(x) = h^{k-j+1} \cdot \frac{1}{(k+1)!} \cdot [\mathbf{M}(x)^{-1} \boldsymbol{\mu}(x)]_j \cdot m^{(k+1)}(x) + o_p(h^{k-j+1}).$$

for  $j = 0, \dots, k$

If  $\mathbf{K}$  is symmetric and  $x \in [h, 1-h]$ ,  $j$  with even  $k-j$

$$[\mathbf{M}(x)^{-1} \boldsymbol{\mu}(x)]_j = [\mathbf{M}^{-1} \boldsymbol{\mu}]_j = \sum_{\ell: j+\ell \text{ even}} [M^{-1}]_{j,\ell} \cdot \mu_{k+1+\ell} = 0$$

Thus,

$$E(\hat{m}^{(j)}(x) \mid \mathbb{X}_n) - m^{(j)}(x) = o_p(h^{k-j+1})$$

# Conditional bias

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If  $p$  has continuous derivative at  $x$  and  $m$  is  $(k + 2)$ -times continuously differentiable at  $x$ ,

$$E\hat{\nu}_\ell(x) = K_h^- * \left( \left( \frac{\cdot - x}{h} \right)^\ell p(\cdot) \right) (x) = \mu_\ell(x)p(x) + \frac{\mu_{\ell+1}(x)}{\ell + 1}p'(x)h + o(h^{\ell+1})$$

$$E\hat{\beta}_\ell(x) = K_h^- * \left( \left( \frac{\cdot - x}{h} \right)^\ell \left( m(\cdot) - \sum_{j=0}^k \frac{(\cdot - x)^j m^{(j)}(x)}{j!} \right) p(\cdot) \right) (x)$$

$$\begin{aligned} &= \frac{\mu_{\ell+k+1}(x)}{(k+1)!} m^{(k+1)}(x)p(x)h^{k+1} + \\ &\frac{\mu_{\ell+k+2}(x)}{(\ell+k+2)!} \left[ \binom{\ell+k+2}{\ell, (k+1), 1} \frac{\ell!}{h^\ell} m^{(k+1)}(x)p'(x)h^{\ell+k+2} + \binom{\ell+k+2}{\ell, (k+2)} \frac{\ell!}{h^\ell} m^{(k+2)}(x)p(x)o(h^{\ell+k+2}) \right] \\ &+ o(h^{k+2}) \end{aligned}$$

# Finding leading term

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$$E(\hat{\mathbf{m}}(x) \mid X_n) - \mathbf{m}(x) = \left( n^{-1} D_x^\top W_x D_x \right)^{-1} n^{-1} D_x^\top W_x (\mathbf{m} - D_x \mathbf{m}(x))$$

$$\left( n^{-1} D_x^\top W_x D_x \right)^{-1} = \left[ \frac{\mathbf{M}^{-1}(x)}{p(x)} - h \left( \frac{\mathbf{M}^{-1}(x)}{p(x)} \right) \left[ \frac{\mu_{i+j+1}(x) p'(x)}{i! j! (i+j+1)} \right]_{i,j} \left( \frac{\mathbf{M}^{-1}(x)}{p(x)} \right) + o(h) \right]$$

$$\frac{1}{n} D_x^\top W_x (\mathbf{m} - D_x \mathbf{m}(x)) = \left[ \frac{p(x)}{(k+1)!} h^{k+1} m^{(k+1)}(x) \boldsymbol{\mu}(x) + C(x) h^{k+2} \left[ \frac{\mu_{i+k+2}(x)}{i!} \right]_i + o(h^{k+2}) \right]$$

where  $C(x) = \frac{m^{(k+1)}(x)}{(k+1)!} p'(x) + \frac{m^{(k+2)}(x)}{(k+2)!} p(x)$

$$E(\hat{\mathbf{m}}(x) \mid X_n) - \mathbf{m}(x) = h^{k+1} \cdot \frac{1}{(k+1)!} \cdot \mathbf{M}(x)^{-1} \boldsymbol{\mu}(x) \cdot m^{(k+1)}(x) + \mathbf{c}(x) h^{k+2} + o_p(h^{k+2}).$$

where  $\mathbf{c}(x) = \frac{\mathbf{M}^{-1}(x)}{p(x)} C(x) \left[ \frac{\mu_{i+k+2}(x)}{i!} \right]_i -$   
 $\left( \frac{\mathbf{M}^{-1}(x)}{p(x)} \right) \left[ \frac{\mu_{i+j+1}(x) p'(x)}{i! j! (i+j+1)} \right]_{i,j} \left( \frac{\mathbf{M}^{-1}(x)}{p(x)} \right) \frac{p(x)}{(k+1)!} h^{k+1} m^{(k+1)}(x) \boldsymbol{\mu}(x)$

# Conditional Variance

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$$\mathbf{S}(x) = \begin{pmatrix} \frac{1}{0!0!} \kappa_0(x) & \frac{1}{0!1!} \kappa_1(x) & \cdots & \frac{1}{0!k!} \kappa_k(x) \\ \frac{1}{1!0!} \kappa_1(x) & \frac{1}{1!1!} \kappa_2(x) & \cdots & \frac{1}{1!k!} \kappa_{k+1}(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{k!0!} \kappa_k(x) & \frac{1}{k!1!} \kappa_{k+1}(x) & \cdots & \frac{1}{k!k!} \kappa_{2k}(x) \end{pmatrix}.$$

Assume that  $\sigma^2$  and  $p$  are continuous at  $x$  and  $p(x) > 0$ . Suppose that  $h \rightarrow 0$  and  $nh \rightarrow \infty$  as  $n \rightarrow \infty$ . Then,

$$\text{var}(\hat{\mathbf{m}}(x) \mid X_1, \dots, X_n) = n^{-1}h^{-1} \cdot \mathbf{M}(x)^{-1} \mathbf{S}(x) \mathbf{M}(x)^{-1} \cdot \frac{\sigma^2(x)}{p(x)} + o_p(n^{-1}h^{-1}).$$

## proof

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Using  $Var[Y_i|\mathbb{X}_n] = Var[Y_i|X_i] = \sigma^2(X_i)$ ,  $Cov[Y_i, Y_j|\mathbb{X}_n] = 0$

$$\text{var}(\hat{m}(x) \mid X_1, \dots, X_n) = \left( \mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x \right)^{-1} \mathbf{D}_x^\top \widetilde{\mathbf{W}}_x \mathbf{D}_x \left( \mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x \right)^{-1}$$

where  $\widetilde{\mathbf{W}}_x := \mathbf{W}_x \text{diag}(\sigma^2(X_i)) \mathbf{W}_x = \text{diag} \left( (K_h(X_i - x))^2 \sigma^2(X_i) \right)$ .

$$n^{-1} \mathbf{D}_x^\top \widetilde{\mathbf{W}}_x \mathbf{D}_x = \frac{1}{n} \left( \frac{\sum_{r=1}^n (X_r - x)^{i+j} (K_h(X_r - X))^2 \sigma^2(X_r)}{i!j!h_{i+j}} \right)_{ij} = \frac{1}{h} \left( \frac{\hat{\gamma}_{i+j}(x)}{i!j!} \right)_{ij}$$

where

$$\hat{\gamma}_\ell(x) := n^{-1} \sum_{i=1}^n \left( \frac{X_i - x}{h} \right)^\ell \left( K_h^2(X_i - x) \sigma^2(X_i) \right), \quad \ell = 0, \dots, 2k$$

$$\begin{aligned} E\hat{\gamma}_\ell(x) &= E \left[ \left( \frac{X_1 - x}{h} \right)^\ell (K^2)_h(X_1 - x) \sigma^2(X_1) \right] \\ &= (K^2)_h^- * \left( \left( \frac{\cdot - x}{h} \right)^\ell \sigma^2(\cdot) p(\cdot) \right) (x) = \frac{\kappa_\ell(x)}{\ell!} \frac{\ell!}{h^\ell} \sigma^2(x) p(x) h^\ell + o(h^\ell) \\ &= \kappa_\ell(x) \sigma^2(x) p(x) + o(h^\ell) \end{aligned}$$

$$\begin{aligned}
 Var \hat{\gamma}_\ell(x) &= \frac{1}{n} Var \left[ \left( \frac{X_1 - x}{h} \right)^\ell (K^2)_h(X_1 - x) \sigma^2(X_1) \right] \\
 &= \frac{1}{nh} (K^4)_h^- * \left( \left( \frac{\cdot - x}{h} \right)^{2\ell} \sigma^4(\cdot) p(\cdot) \right) (x) + o\left(\frac{1}{nh}\right) \\
 &= \frac{1}{nh} \left( \frac{\int_0^1 \left( \frac{u-x}{h} \right)^{2\ell} (K^4)_h(u-x) du}{(2\ell)!} (2\ell)! \sigma^4(x) p(x) + o(h^{2\ell}) \right) + o\left(\frac{1}{nh}\right) \\
 O_p(\sqrt{Var}) &= O_p\left(\frac{1}{\sqrt{nh}}\right) = o_p(1)
 \end{aligned}$$

$$\hat{\gamma}_\ell(x) = \kappa_\ell(x)\sigma^2(x)p(x) + o(h^\ell) + o_p(1) = \kappa_\ell(x)\sigma^2(x)p(x) + o_p(1)$$

$$n^{-1}\mathbf{D}_x^\top \widetilde{\mathbf{W}}_x \mathbf{D}_x = \frac{1}{h} \left( \frac{\hat{\gamma}_{i+j}(x)}{i!j!} \right)_{ij} = \frac{\sigma^2(x)p(x)}{h} (\mathbf{S}(x) + o_p(1))$$

$$\begin{aligned}
 \text{var}(\hat{\mathbf{m}}(x) \mid X_1, \dots, X_n) &= \left( \mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x \right)^{-1} \mathbf{D}_x^\top \widetilde{\mathbf{W}}_x \mathbf{D}_x \left( \mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x \right)^{-1} \\
 &= \frac{1}{n} \left( \frac{1}{p(x)} [\mathbf{M}(x)]^{-1} + o_p(1) \right) \frac{\sigma^2(x)p(x)}{h} (\mathbf{S}(x) + o_p(1)) \left( \frac{1}{p(x)} [\mathbf{M}(x)]^{-1} + o_p(1) \right) \\
 &= n^{-1}h^{-1} \cdot \mathbf{M}(x)^{-1} \mathbf{S}(x) \mathbf{M}(x)^{-1} \cdot \frac{\sigma^2(x)}{p(x)} + o_p(n^{-1}h^{-1})
 \end{aligned}$$

## j-th derivative

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$$\text{var}(\hat{m}^{(j)}(x) \mid X_1, \dots, X_n) = n^{-1} h^{-2j-1} [\mathbf{M}(x)^{-1} \mathbf{S}(x) \mathbf{M}(x)^{-1}]_{j,j} \frac{\sigma^2(x)}{p(x)} + o_p(n^{-1} h^{-2j-1})$$

j with even k-j,  $x \in [h, 1-h]$ , all odd moments of  $K, K^2$  vanish

$$\begin{aligned} [\mathbf{M}^{-1} \mathbf{S} \mathbf{M}^{-1}]_{j,j} &= \sum_{r=0}^k \sum_{l=0}^k (\mathbf{M}^{-1})_{j,r} (\mathbf{S})_{r,l} (\mathbf{M}^{-1})_{r,j} \\ &= \sum_{r:r-j \text{ even}}^k \sum_{l:l-j \text{ even}}^k (\mathbf{M}^{-1})_{j,r} (\mathbf{S})_{r,l} (\mathbf{M}^{-1})_{r,j} \\ &= \sum_{r:r-j \text{ even}}^{k+1} \sum_{l:l-j \text{ even}}^{k+1} (\mathbf{M}^{-1})_{j,r} (\mathbf{S})_{r,l} (\mathbf{M}^{-1})_{r,j} \end{aligned}$$

coefficient of  $n^{-1} h^{-2j-1}$  for the k-th order local polynomial estimator is identical to that of (k+1)-th order local polynomial estimator.

# Invertibility of $\mathbf{M}(x)$

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For  $\lambda \in \mathbb{R}^{k+1}$ ,

$$\lambda^\top \mathbf{M}(x) \lambda = \int_{-x/h}^{(1-x)/h} K(v) \cdot \left( \lambda_0 \frac{1}{0!} + \lambda_1 \frac{v}{1!} + \dots + \lambda_k \frac{v^k}{k!} \right)^2 dv.$$

Suppose that  $\lambda^\top \mathbf{M}(x) \lambda = 0$ . Then,

$$\text{Leb} \left( \left\{ v \in [-x/h, (1-x)/h] : K(v) > 0, \left( \lambda_0 \frac{1}{0!} + \lambda_1 \frac{v}{1!} + \dots + \lambda_k \frac{v^k}{k!} \right) \neq 0 \right\} \right) = 0.$$

Since  $K \geq 0$  is supported on  $[-1, 1]$ , we can conclude  $\lambda_0 = \lambda_1 = \dots = \lambda_k = 0$ .

If support of  $K$  contains non-empty interior on both  $[-1, 0]$  and  $[0, 1]$ ,  $\mathbf{M}(x)$  is invertible since  $[-x/h, (1-x)/h]$  contains  $[0, 1]$  or  $[-1, 0]$ .

# Pointwise rate of Convergence

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Assume that  $m$  is  $(k + 1)$ -times continuously differentiable at  $x$  and  $\sigma^2, p$  are continuous at  $x$  and  $p(x) > 0$ . Suppose that  $h \rightarrow 0$  and  $nh \rightarrow \infty$  as  $n \rightarrow \infty$

$$\|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\| = O_p(h^{k+1} + n^{-1/2}h^{-1/2})$$

Using

$$\begin{aligned}\mathbb{E} \left( \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 \mid \mathbb{X}_n \right) &= (\mathbb{E}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) - \mathbf{m}(x))^T (\mathbb{E}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) - \mathbf{m}(x)) \\ &\quad + \text{tr}(\text{var}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n)) \\ &= \left\| h^{k+1} \frac{1}{(k+1)!} \mathbf{M}(x)^{-1} \boldsymbol{\mu}(x) m^{(k+1)}(x) + o_p(h^{k+1}) \right\|^2 \\ &\quad + \frac{1}{nh} \frac{\sigma^2(x)}{p(x)} \text{tr}(\mathbf{M}(x)^{-1} \mathbf{S}(x) \mathbf{M}(x)^{-1}) + o_p\left(\frac{1}{nh}\right)\end{aligned}$$

# proof

---

For fixed  $x \in (0, 1)$ ,

$\mathbf{M}(x) = \mathbf{M}, \boldsymbol{\mu}(x) = \boldsymbol{\mu}, \mathbf{S}(x) = \mathbf{S}$  for large  $n$

since  $\mu_j(x) = \int_{-x/h}^{(1-x)/h} v^j K(v) dv, \kappa_j(x) = \int_{-x/h}^{(1-x)/h} v^j K^2(v) dv$

Therefore,

$$\mathbb{E} \left( \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 \mid \mathbb{X}_n \right) \leq \left[ h^{k+1} a + o_p(h^{k+1}) \right]^T \left[ h^{k+1} a + o_p(h^{k+1}) \right] + \frac{b}{nh} + o_p\left(\frac{1}{nh}\right)$$

for some  $a, b \in \mathbb{R}$ .

$$= h^{2k+2} a^T a + o_p(h^{2k+2}) + \frac{b}{nh} + o_p\left(\frac{1}{nh}\right)$$

$$Z_n = ||\hat{\mathbf{m}}(x) - \mathbf{m}(x)||^2$$

$$E \left[ \frac{|Z_n|}{(h^{k+1} + \frac{1}{\sqrt{nh}})^2} \mid \mathbb{X}_n \right] \leq 2\max(a^T a, b) + o_p(1) = C + o_p(1)$$

$$P \left( E \left[ \frac{|Z_n|}{(h^{k+1} + \frac{1}{\sqrt{nh}})^2} \mid \mathbb{X}_n \right] < C + 1 \right) \rightarrow 1$$

$\mathcal{E}_n : D_x^T W_w D_x$  is invertible,  $P(\mathcal{E}_n) \rightarrow 1$  Therefore,

$$||\hat{\mathbf{m}}(x) - \mathbf{m}(x)|| = O_p(h^{k+1} + n^{-1/2}h^{-1/2})$$

## Conditional expected $L^2$ error

---

Assume that  $m$  has bounded  $(k+1)$ th derivative on  $[0, 1]$ , that  $p$  satisfies

$$0 < \inf_{x \in [0,1]} p(x) \leq \sup_{x \in [0,1]} p(x) < \infty$$

and that  $\sigma^2$  is *bounded* on  $[0, 1]$ . Suppose that  $K$  is Lipschitz continuous and that  $h \rightarrow 0$  and

$$(nh)/\log n \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

Then,

$$\mathbb{E} \left( \int_0^1 \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 dx \mid X_1, \dots, X_n \right) = O_p \left( h^{2(k+1)} + n^{-1} h^{-1} \right).$$

First, we prove

$$\hat{\lambda}_{\inf} := \inf_{x \in [0,1]} \lambda_{\min}(n^{-1} \mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x) > \varepsilon$$

with probability tending to 1.

**Hoffman-Wielandt inequality.** Let  $\mathbf{A}$  and  $\mathbf{B}$  be real symmetric matrices. Let  $\lambda_j$  and  $\eta_j$  be the ordered eigenvalues of  $\mathbf{A}$  and  $\mathbf{B}$ , respectively. Then,  $\sum_j |\lambda_j - \eta_j|^2 \leq \text{tr}[(\mathbf{A} - \mathbf{B})^2]$ .

The Hoffman-Wielandt inequality implies

$$\lambda_{\min}(\mathbf{A}) \geq \lambda_{\min}(\mathbf{B}) - \sqrt{\text{tr}[(\mathbf{A} - \mathbf{B})^2]}. \quad (7.11)$$

Put

$$\varepsilon = \frac{1}{2} \cdot \inf_{x \in [0,1]} p(x) \cdot \inf_{x \in [0,1]} \lambda_{\min}(\mathbf{M}(x))$$

Applying (7.11), we get

$$\hat{\lambda}_{\inf} \geq 2\varepsilon - \sup_{x \in [0,1]} \sqrt{\text{tr} \left[ \left( n^{-1} \mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x - \mathbf{M}(x) \cdot p(x) \right)^2 \right]}.$$

It remains to prove  $\varepsilon > 0$ , for which it suffices to show

$$\inf_{x \in [0,1]} \lambda_{\min}(\mathbf{M}(x)) > 0. \quad (7.12)$$

Note that

$$\inf_{x \in [h, 1-h]} \lambda_{\min}(\mathbf{M}(x)) = \lambda_{\min} \left( \begin{pmatrix} 1 & \mu_1 \\ \mu_1 & \mu_2 \end{pmatrix} \right) > 0. \quad (7.13)$$

Now, we may write  $x \in [0, h]$  as  $x = \alpha \cdot h$  with  $\alpha \in [0, 1]$ . Thus, for sufficiently large  $n$ ,

## proof

---

In case of  $x = \alpha h$  with  $\alpha \in [0, 1]$

$$|f(\alpha + \delta) - f(\alpha)|^2 \leq \text{tr} \left[ \left( \int_{-\alpha-\delta}^{\alpha} \frac{v^{i+j} K(v)}{i!j!} \right)_{i,j}^2 \right] \rightarrow 0 \text{ as } \delta \rightarrow 0$$

In case of  $x = 1 - \alpha h$  with  $\alpha \in [0, 1]$

$$|f(\alpha + \delta) - f(\alpha)|^2 \leq \text{tr} \left[ \left( \int_{\alpha}^{\alpha+\delta} \frac{v^{i+j} K(v)}{i!j!} \right)_{i,j}^2 \right] \rightarrow 0 \text{ as } \delta \rightarrow 0$$

$$\begin{aligned} \inf_{x \in [0, h]} \lambda_{\min}(\mathbf{M}(x)) &= \inf_{\alpha \in [0, 1]} \lambda_{\min} \left( \begin{pmatrix} \int_{-\alpha}^1 K(v) dv & \int_{-\alpha}^1 v K(v) dv \\ \int_{-\alpha}^1 v K(v) dv & \int_{-\alpha}^1 v^2 K(v) dv \end{pmatrix} \right) \\ &=: \inf_{\alpha \in [0, 1]} f(\alpha). \end{aligned}$$

The function  $f$  on  $[0, 1]$  is continuous, which follows from another application of the Hoffman-Wielandt inequality. Also,  $f(\alpha) > 0$  for all  $\alpha \in [0, 1]$ . This proves  $\inf_{\alpha \in [0, 1]} f(\alpha) > 0$ . The proof of

# proof

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Next, we prove

$$\sup_{x \in [0,1]} |\hat{\beta}_\ell(x)| \leq C \cdot h^{k+1}, \quad \ell = 0, \dots, k$$

with probability tending to 1.

Using

$$\hat{\beta}_\ell(x) := n^{-1} \sum_{i=1}^n \left( \frac{X_i - x}{h} \right)^\ell K_h(X_i - x) \left( m(X_i) - \sum_{j=0}^k \frac{(X_i - x)^j m^{(j)}(x)}{j!} \right), \ell = 0, \dots, k$$

$$m(X_i) - \sum_{j=0}^k \frac{(X_i - x)^j m^{(j)}(x)}{j!} = \frac{m^{(k+1)}(\tilde{x}_i)}{(k+1)!} (X_i - x)^{k+1}$$

## proof

---

with  $\tilde{K}_h(X_i - x) = \left| \frac{X_i - x}{h} \right|^\ell \cdot K_h(X_i - x)$ ,

$$\begin{aligned} \sup_{x \in [0,1]} |\hat{\beta}_\ell(x)| &\leq \frac{h^{k+1}}{(k+1)!} \cdot \sup_{x \in [0,1]} |m^{(k+1)}(x)| \cdot \left[ \sup_{x \in [0,1]} \mathbb{E} \left( \tilde{K}_h(X_1 - x) \right) \right. \\ &\quad \left. + \sup_{x \in [0,1]} \left| n^{-1} \sum_{i=1}^n \tilde{K}_h(X_i - x) - \mathbb{E} \left( \tilde{K}_h(X_1 - x) \right) \right| \right] \end{aligned}$$

also, since  $p$  is bounded,  $\sup_{x \in [0,1]} |K_h^- * p(x) - \mu_0(x)p(x)| \leq C$  for large  $n$ .

$\tilde{K}(v) = v^l K(v)$  is still Lipschitz conti since

$$|\tilde{K}(x) - \tilde{K}(y)| \leq |x|^l |K(x) - K(y)| + |x^l - y^l| |K(y)| \leq C_1 |x - y| + l C_2 |x - y|$$

Then we get

$$\leq O(h^{k+1}) + O_p(h^{k+1} n^{-1/2} h^{-1/2} \sqrt{\log n}) = O(h^{k+1}) + o_p(h^{k+1})$$

## proof

---

Next, we prove

$$\sup_{x \in [0,1]} |\hat{\gamma}_\ell(x)| \leq C, \quad \ell = 0, \dots, 2k$$

with probability tending to 1.

$$\begin{aligned} \sup_{x \in [0,1]} |\hat{\gamma}_\ell(x)| &\leq \sup_{x \in [0,1]} \mathbb{E} \left( \left( \frac{X_1 - x}{h} \right)^\ell (K^2)_h(X_1 - x) \sigma^2(X_1) \right) \\ &+ \sup_{x \in [0,1]} \left| \frac{1}{n} \sum_{i=1}^n \left( \frac{X_i - x}{h} \right)^\ell (K^2)_h(X_i - x) \sigma^2(X_i) - \mathbb{E} \left( \left( \frac{X_1 - x}{h} \right)^\ell (K^2)_h(X_1 - x) \sigma^2(X_1) \right) \right| \end{aligned}$$

and we know that

$$\sup_{x \in [0,1]} \left| \mathbb{E} \left( \left( \frac{X_1 - x}{h} \right)^\ell (K^2)_h(X_1 - x) \sigma^2(X_1) \right) - \frac{\kappa_{2\ell}(x)}{\ell!} \ell! \sigma^2(x) p(x) \right| \leq Ch^0$$

$$\sup_{x \in [0,1]} \left| \frac{1}{n} \sum_{i=1}^n \left( \frac{X_i - x}{h} \right)^\ell (K^2)_h(X_i - x) \sigma^2(X_i) - \mathbb{E} \left( \left( \frac{X_1 - x}{h} \right)^\ell (K^2)_h(X_1 - x) \sigma^2(X_1) \right) \right|$$

$$= O_p(n^{-1/2} h^{-1/2} \sqrt{\log n})$$

since with  $\xi_{ni}(x) = K_h(X_i - x) \sigma^2(X_i) \left( \frac{X_i - x}{h} \right)^\ell - E[K_h(X_1 - x) \sigma^2(X_1) \left( \frac{X_1 - x}{h} \right)^\ell]$ ,

$E\xi_{ni}(x) = 0$ ,  $E\xi_{ni}^2(x) = \frac{1}{h} (K^2)_h * (\cdot - x)^\ell \sigma^2(\cdot) p(\cdot)(x) \frac{1}{h^\ell} \leq C \frac{1}{h}$  Also,

$$\sup_{|x-y| \leq n^{-\gamma}} |Z_n(x) - Z_n(y)| \leq \frac{C}{nh} \sup_{|x-y| \leq n^{-\gamma}} \sum_i [K\left(\frac{X_i - x}{h}\right) \left(\frac{X_i - x}{h}\right)^l - K\left(\frac{X_i - y}{h}\right) \left(\frac{X_i - y}{h}\right)^l] + C' \sup_{|x-y| \leq n^{-\gamma}} E[K\left(\frac{X_1 - x}{h}\right) \left(\frac{X_1 - x}{h}\right)^l - K\left(\frac{X_1 - y}{h}\right) \left(\frac{X_1 - y}{h}\right)^l] \leq C'' \frac{n^{-\gamma}}{h^2}$$

$$\begin{aligned}
 \sup_{x \in [0,1]} \mathbb{E} \left( \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 \mid \mathbb{X}_n \right) &\leq \sup_{x \in [0,1]} \left( \mathbb{E}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) - \mathbf{m}(x) \right)^T \left( \mathbb{E}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) - \mathbf{m}(x) \right) \\
 &\quad + \sup_{x \in [0,1]} \text{tr} \left( \text{var}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) \right) \\
 &\leq \sup_{x \in [0,1]} \left\| \left( \mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x \right)^{-1} D_x^T W_x (\mathbf{m} - D_x \mathbf{m}(x)) \right\|^2 \\
 &\quad + \sup_{x \in [0,1]} \text{tr} \left( \left( \mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x \right)^{-1} \mathbf{D}_x^\top \widetilde{\mathbf{W}}_x \mathbf{D}_x \left( \mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x \right)^{-1} \right)
 \end{aligned}$$

Using for a real  $k \times k$  symmetric matrix  $\mathbf{A}$  with eigenvalues  $\eta_j$  and a nonnegative definite symmetric  $k \times k$  matrix  $\mathbf{B} = (B_{ij})$ ,

$$\text{tr}(\mathbf{A}\mathbf{B}\mathbf{A}) \leq \text{tr}(\mathbf{B}) \cdot \max_j \eta_j^2.$$

$$\|\mathbf{A}\mathbf{b}\|^2 \leq \|\mathbf{b}\|^2 \cdot \max_j \eta_j^2.$$

$$\begin{aligned}
 &\leq \hat{\lambda}_{\inf}^{-2} \cdot \sup_{x \in [0,1]} \left( |\hat{\beta}_0(x)|^2 + |\hat{\beta}_1(x)|^2 + \dots + |\hat{\beta}_k(x)|^2 \right) \\
 &\quad + \hat{\lambda}_{\inf}^{-2} \cdot \sup_{x \in [0,1]} (\hat{\gamma}_0(x) + \hat{\gamma}_2(x) + \dots + \hat{\gamma}_{2k}(x)) \cdot n^{-1} h^{-1} \\
 &\leq C' \left( h^{2k+2} + n^{-1} h^{-1} \right)
 \end{aligned}$$

with probability tending to 1. for any constant  $C'$  such that  $\frac{(k+1)C(C \vee 1)}{\epsilon^2} < C' < \infty$

Assume that  $m$  has bounded  $(k+1)$ th derivative on  $[0, 1]$ , that  $p$  satisfies

$$0 < \inf_{x \in [0,1]} p(x) \leq \sup_{x \in [0,1]} p(x) < \infty$$

and that  $\sigma^2$  is *bounded* on  $[0, 1]$ . Suppose that  $K$  is Lipschitz continuous and that  $h \rightarrow 0$  and

$$(nh)/\log n \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

$$\int_0^1 \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 dx = O_p \left( h^{2(k+1)} + n^{-1} h^{-1} \right).$$

## proof of $L^2$ error

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We have proved that

$$\mathbb{E} \left( \int_0^1 \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 dx \mid X_1, \dots, X_n \right) \leq C' (h^{2k+2} + n^{-1}h^{-1})$$

for some  $C' < \infty$ , with probability tending to 1.

Let  $Z_n = \int_0^1 \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 dx$ , Then

$$P \left( E \left[ \frac{|Z_n|}{(h^{2k+2} + n^{-1}h^{-1})} \mid \mathbb{X}_n \right] \leq C' \right) \rightarrow 1$$

$\mathcal{E}_n : \inf_{x \in [0,1]} \lambda_{\min}(\frac{1}{n} D_x^T W_x D_x) > \epsilon$ ,  $P(\mathcal{E}_n) \rightarrow 1$  Therefore,

$$\int_0^1 \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 dx = O_p(h^{2k+2} + \frac{1}{nh})$$

## j-th derivative

---

$$\int_0^1 \left\| \hat{m}^{(j)}(x) - m^{(j)}(x) \right\|^2 dx = O_p \left( h^{2(k-j+1)} + n^{-1} h^{-2j-1} \right), \quad 0 \leq j \leq k.$$

If  $K$  is symmetric and  $x \in [h, 1-h]$ ,  $j$  with even  $k-j$ ,  $p$  has continuous derivative at  $x$  and  $m$  is  $(k+2)$ -times continuously differentiable at  $x$ ,

Recall

$$\hat{\beta}_\ell(x) := n^{-1} \sum_{i=1}^n \left( \frac{X_i - x}{h} \right)^\ell K_h(X_i - x) \left( m(X_i) - \sum_{j=0}^k \frac{(X_i - x)^j m^{(j)}(x)}{j!} \right), \quad \ell = 0, \dots, k$$

$$m(X_i) - \sum_{j=0}^k \frac{(X_i - x)^j m^{(j)}(x)}{j!} = \frac{m^{(k+1)}(x)}{(k+1)!} (X_i - x)^{k+1} + \frac{m^{(k+2)}(\tilde{x}_i)}{(k+2)!} (X_i - x)^{k+2}$$

## j-th derivative

---

$$\begin{aligned}
 \sup_{x \in [h, 1-h]} |\hat{\beta}_j(x)| &\leq \frac{\sup_{x \in [h, 1-h]} |m^{(k+1)}(x)|}{(k+1)!} h^{k+1} \cdot \left[ \sup_{x \in [h, 1-h]} \mathbb{E} \left( K_h(X_1 - x) \left( \frac{X_1 - x}{h} \right)^{j+k+1} \right) \right. \\
 &+ \left. \sup_{x \in [h, 1-h]} \left| n^{-1} \sum_{i=1}^n K_h(X_i - x) \left( \frac{X_i - x}{h} \right)^{j+k+1} - \mathbb{E} \left( K_h(X_1 - x) \left( \frac{X_1 - x}{h} \right)^{j+k+1} \right) \right| \right] \\
 &+ \frac{\sup_{x \in [0, 1]} |m^{(k+2)}(x)|}{(k+2)!} h^{k+2} \cdot \left[ \sup_{x \in [h, 1-h]} \mathbb{E} \left( K_h(X_1 - x) \left( \frac{X_1 - x}{h} \right)^j \right) \right. \\
 &+ \left. \sup_{x \in [h, 1-h]} \left| n^{-1} \sum_{i=1}^n K_h(X_i - x) \left( \frac{X_i - x}{h} \right)^j - \mathbb{E} \left( K_h(X_1 - x) \left( \frac{X_1 - x}{h} \right)^j \right) \right| \right] \\
 &\leq C' (h^{k+2} + \frac{1}{\sqrt{nh}})
 \end{aligned}$$

with probability tending to 1.

## j-th derivative

---

$$\begin{aligned} & \mathbb{E} \left( K_h(X_1 - x) \left( \frac{X_1 - x}{h} \right)^{j+k+1} \right) \\ &= \frac{1}{h^{j+k+1}} K_h^- * (\cdot - x)^{j+k+1} p(\cdot)(x) = \mu_{j+k+1} p(x) + \mu_{j+k+2} p'(x) h + o_p(h) \\ &= \mu_{j+k+2} p'(x) h + o_p(h) \\ & \sup_{x \in [0,1]} \left| n^{-1} \sum_{i=1}^n \tilde{K}(X_i - x) - \mathbb{E}(\tilde{K}(X_1 - x)) \right| = O_p \left( n^{-1/2} h^{-1/2} \sqrt{\log n} \right). \end{aligned}$$

where  $\tilde{K}(v) = v^l K(v)$  which is still Lipschitz conti

## j-th derivative

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Using

$$\sup_{x \in [h, 1-h]} E(\|\hat{m}^{(j)}(x) - m^{(j)}(x)\|^2 | \mathbb{X}_n) \leq \hat{\lambda}_{inf}^{-2} \sup_{x \in [h, 1-h]} |\hat{\beta}_j(x)|^2 + \hat{\lambda}_{inf}^{-2} \sup_{x \in [h, 1-h]} |\hat{\gamma}_{2j}(x)|^2 \frac{1}{nh}$$

we get

$$\begin{aligned} \int_h^{1-h} \left\| \hat{m}^{(j)}(x) - m^{(j)}(x) \right\|^2 dx &= O_p \left( h^{2k-2j+4} + n^{-1} h^{-2j-1} \right) \\ \int_0^1 \left\| \hat{m}^{(j)}(x) - m^{(j)}(x) \right\|^2 dx &= (1-2h) O_p \left( h^{2k-2j+4} + \frac{1}{nh^{2j+1}} \right) + 2h O_p \left( h^{2k-2j+2} + \frac{1}{nh^{2j+1}} \right) \\ &= O_p \left( h^{2k-2j+3} + \frac{1}{nh^{2j+1}} \right) \end{aligned}$$

Assume that  $E[\exp(\alpha|Y - m(X)|)] < \infty$  for some  $\alpha > 0$ ,  $E(Y^2|X = \cdot)$  is bounded on  $[0, 1]$  and  $0 < \inf_{x \in [0, 1]} \rho(x) < \sup_{x \in [0, 1]} \rho(x) < \infty$ .

Suppose that  $K$  is Lipschitz continuous and that  $\frac{nh}{(\log n)^3} \rightarrow \infty$  as  $n \rightarrow \infty$ . Then, it holds that

$$\sup_{x \in [0, 1]} \|\hat{m}(x) - \mathbb{E}(\hat{m}(x)|X_1, \dots, X_n)\| = O_p\left(n^{-1/2}h^{-1/2}\sqrt{\log n}\right).$$

$$\hat{\mathbf{m}}(x) - \mathbb{E}(\hat{\mathbf{m}}(x)|\mathbb{X}_n) = (D_x^\top W_x D_x)^{-1} D_x^\top W_x (Y - \mathbf{m}).$$

$$\sup_{x \in [0,1]} \|\hat{\mathbf{m}}(x) - \mathbb{E}(\hat{\mathbf{m}}(x)|\mathbb{X}_n)\| \leq \hat{\lambda}_{\inf}^{-1} \sup_{x \in [0,1]} \left\| n^{-1} D_x^\top W_x (Y - \mathbf{m}) \right\|.$$

$$\sup_{x \in [0,1]} \left| n^{-1} \sum_{i=1}^n \left( \frac{X_i - x}{h} \right)^\ell K_h(X_i - x) (Y_i - m(X_i)) \right| = O_p \left( n^{-1/2} h^{-1/2} \sqrt{\log n} \right), \ell = 0, \dots, k$$

Lemma 6.1, with  $\epsilon_{ni} = Y_i - m(X_i)$ ,  $\tilde{K}(v) = v^l K(v)$

## Lemma 6.1

For iid copies  $(X_i, \epsilon_{ni})$ ,  $1 \leq i \leq n$ , with  $\mathbb{E}(\epsilon_{ni}|X_i) = 0$ . Assume that  $\sup_n \mathbb{E}[\exp(\alpha|\epsilon_{ni}|)] < \infty$  for some  $\alpha > 0$ ,  $\sup_n \mathbb{E}(\epsilon_{ni}^2|X_i = \cdot)$  and  $p$  are bounded on  $[0, 1]$ . Suppose that  $K$  is Lipschitz continuous and  $(nh)/(\log n)^3 \rightarrow \infty$  as  $n \rightarrow \infty$ . Then, it holds that

$$\sup_{x \in [0, 1]} \left| n^{-1} \sum_{i=1}^n K_h(X_i - x) \cdot \epsilon_{ni} \right| = O_p \left( n^{-1/2} h^{-1/2} \sqrt{\log n} \right).$$

# $L^\infty$ error proof

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**Proof.** Put  $\tau_n = \sqrt{nh/\log n}$ . Define

$$Z_n(x) = n^{-1} \sum_{i=1}^n K_h(X_i - x) \left[ \varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) - \mathbb{E}(\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) | X_i) \right].$$

There exists a constant  $0 < C < \infty$  such that

$$\begin{aligned} \mathbb{P}\left(\max_{1 \leq i \leq n} |\varepsilon_{ni}| \leq \tau_n\right) &\geq 1 - C \cdot n \cdot \exp(-\alpha \tau_n) \\ &= 1 - C \cdot \exp(\log n - \alpha \tau_n) \rightarrow 1. \end{aligned}$$

This shows that

$$\sup_{x \in [0,1]} \left| n^{-1} \sum_{i=1}^n K_h(X_i - x) \varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| > \tau_n) \right| = 0 \quad (6.1)$$

with probability tending to one.

## $L^\infty$ error proof

---

Furthermore, we get that

$$\begin{aligned} & \sup_{x \in [0,1]} \left| n^{-1} \sum_{i=1}^n K_h(X_i - x) \cdot \mathbb{E}(\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| > \tau_n) | X_i) \right| \\ & \leq h^{-1} \cdot (\sup K) \cdot n^{-1} \sum_{i=1}^n \mathbb{E}(|\varepsilon_{ni}| \cdot \mathbb{I}(|\varepsilon_{ni}| > \tau_n) | X_i) \\ & \leq h^{-1} \cdot e^{-\alpha \tau_n / 2} \cdot (\sup K) \cdot \frac{2}{\alpha} \cdot n^{-1} \sum_{i=1}^n \mathbb{E} \left( e^{\alpha |\varepsilon_{ni}|} | X_i \right) \\ & \leq C \cdot (nh)^{-1/2} \exp \left( \frac{1}{2} \log \frac{n}{h} - \frac{\alpha}{2} \sqrt{\frac{nh}{\log n}} \right) \\ & \leq C \cdot (nh)^{-1/2} \exp \left( \log n - \frac{\alpha}{2} \sqrt{\frac{nh}{\log n}} \right), \end{aligned}$$

with the 3rd inequality holding with probability tending to one.

The 4th inequality follows since  $nh \gg 1$ . Since  $nh/(\log n)^3 \rightarrow \infty$ , the RHS of the fourth inequality equals  $o(n^{-1/2}h^{-1/2})$ . This shows

$$\sup_{x \in [0,1]} \left| n^{-1} \sum_{i=1}^n K_h(X_i - x) \cdot \mathbb{E}(\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| > \tau_n) | X_i) \right| = o_p(n^{-1/2}h^{-1/2}),$$

which together with (6.1) entails

$$\sup_{x \in [0,1]} \left| n^{-1} \sum_{i=1}^n K_h(X_i - x) \cdot \varepsilon_{ni} \right| = \sup_{x \in [0,1]} |Z_n(x)| + o_p(n^{-1/2}h^{-1/2}).$$

Now, let  $\mathcal{N}(\delta)$  be  $\delta$ -net such that, for any  $x \in [0, 1]$ , there exists a point  $x' \in \mathcal{N}(\delta)$  such that  $|x - x'| \leq \delta$ .

# $L^\infty$ error proof

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$$|Z_n(x) - Z_n(x')| \leq \frac{1}{nh} \sum |K(\frac{X_i - x}{h})(\frac{X_i - x}{h})^l - K(\frac{X_i - x'}{h})(\frac{X_i - x'}{h})^l| [W_{ni}]$$

We claim

$$\sup_{x \in \mathcal{N}(n^{-\gamma})} |Z_n(x)| = O_p(\tau_n^{-1}) \quad (6.2)$$

for any  $\gamma > 0$ . This proves the lemma since

$$\sup_{x, x' \in [0,1], |x-x'| \leq n^{-\gamma}} |Z_n(x) - Z_n(x')| \leq C \cdot h^{-1} \cdot h^{-1} \cdot n^{-\gamma} \cdot n^{-1} \sum_{i=1}^n W_i$$

for some iid  $W_i$  with finite moment. The claim (6.2) follows if we prove

$$\sup_{x \in [0,1]} \mathbb{P}(|Z_n(x)| > C \cdot \tau_n^{-1}) \leq C_0 \cdot n^{-g(C)} \quad (6.3)$$

for some constant  $0 < C_0 < \infty$  and some function  $g$  such that  $g(C) \uparrow \infty$  as  $C \uparrow \infty$ . This is because

$$\mathbb{P}\left(\sup_{x \in \mathcal{N}(n^{-\gamma})} |Z_n(x)| > C \cdot \tau_n^{-1}\right) \leq \frac{1}{2} \cdot n^\gamma \cdot \sup_{x \in [0,1]} \mathbb{P}(|Z_n(x)| > C \cdot \tau_n^{-1}).$$

$$\xi_{ni}(x) = K_h(X_i - x) \left( \frac{X_i - x}{h} \right)^l [\dots]$$

Proof of (6.3): Put

$$\xi_{ni}(x) = K_h(X_i - x) \left( \varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) - \mathbb{E}(\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) | X_i) \right).$$

Note that

$$\mathbb{P}(|Z_n(x)| > C \cdot \tau_n^{-1}) = \mathbb{P}(Z_n(x) > C \cdot \tau_n^{-1}) + \mathbb{P}(-Z_n(x) > C \cdot \tau_n^{-1}).$$

We find a bound to the first term. The second one can be treated in the same way. We have

$$\begin{aligned} \mathbb{P}(Z_n(x) > C \cdot \tau_n^{-1}) &= \mathbb{P}\left(\sqrt{\frac{h \log n}{n}} \sum_{i=1}^n \xi_{ni}(x) > C \log n\right) \\ &\leq n^{-C} \left( \mathbb{E} \exp\left(\sqrt{\frac{h \log n}{n}} \xi_{n1}(x)\right) \right)^n. \end{aligned} \quad (6.4)$$

## $L^\infty$ error proof

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$$\begin{aligned} E\xi_{ni}(x) &= 0, E(\xi_{ni}(x)^2) \leq 4h^{-1}E[(K^2)_h(X_1 - x)\left(\frac{X_1 - x}{h}\right)^l \epsilon_{ni}^2] \\ &\leq h^{-1}C \cdot E[(K^2)_h(X_1 - x)\left(\frac{X_1 - x}{h}\right)^l] \leq h^{-1}C' \end{aligned}$$

Further elaborating on the RHS of (6.4),

$$\begin{aligned} &n^{-C} \left( \mathbb{E} \exp \left( \sqrt{\frac{h \log n}{n}} \xi_{n1}(x) \right) \right)^n \\ &\leq n^{-C} \left( 1 + e^{2(\sup K)} \cdot \frac{h \log n}{2n} \mathbb{E}(\xi_{n1}(x)^2) \right)^n \\ &\leq n^{-C} \exp \left( e^{2(\sup K)} \cdot n \left( \frac{h \log n}{2n} \right) \mathbb{E}(\xi_{n1}(x)^2) \right), \end{aligned}$$

where we have used  $e^a = 1 + a + (a^2/2)e^{a'}$  for some  $a' : |a'| \leq |a|$  for the first inequality and  $(1 + \lambda)^n \leq e^{n\lambda}$  for  $\lambda > -1$  for the second. The claim (6.3) now follows from  $\mathbb{E}(\xi_{n1}(x)^2) \leq 4h^{-1}\mathbb{E}[(K^2)_h(X_1 - x)\epsilon_{n1}^2] \leq C'h^{-1}$  for sufficiently large  $n$ , where  $0 < C' < \infty$  is an absolute constant.  $\square$

Under same assumption,

$$\sup_{x \in [0,1]} \|\hat{m}(x) - m(x)\| = O_p \left( h^{k+1} + n^{-1/2} h^{-1/2} \sqrt{\log n} \right).$$

since

$$\sup_{x \in [0,1]} \|\hat{\mathbf{m}}(x) - \mathbb{E}(\hat{\mathbf{m}}(x) | X_1, \dots, X_n)\| = O_p \left( n^{-1/2} h^{-1/2} \sqrt{\log n} \right).$$

$$\sup_{x \in [0,1]} \|E(\hat{\mathbf{m}}(x) | \mathbb{X}_n) - \mathbf{m}(x)\|^2 \leq \hat{\lambda}_{inf}^{-2} \sup_{x \in [0,1]} \sum_{l=0}^k |\hat{\beta}_l(x)|^2 \leq C(h^{k+1})^2$$

with probability tending to 1

## j-th derivative

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$$\sup_{x \in [0,1]} \left\| \hat{m}^{(j)}(x) - m^{(j)}(x) \right\| = O_p \left( h^{k-j+1} + n^{-1/2} h^{-j-1/2} \sqrt{\log n} \right) \quad j = 0, \dots, k$$

Also, using

$$\sup_{x \in [h, 1-h]} \left| \hat{\beta}_j(x) \right| \leq C' (h^{k+2} + \frac{1}{\sqrt{nh}})$$

We get that

$$\sup_{x \in [h, 1-h]} \left\| \hat{m}^{(j)}(x) - m^{(j)}(x) \right\| = O_p \left( h^{k-j+2} + n^{-1/2} h^{-j-1/2} \sqrt{\log n} \right),$$

under the additional assumption that  $p$  has bounded derivative and  $m$  has bounded  $(k+2)$ -th derivative on  $[0, 1]$  in case of even  $k-j$ , symmetric K

# Asymptotic distribution

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## Theorem 9.6

Let  $x \in [0, 1]$  be fixed. Assume that  $\mathbb{E}(|Y - m(X)|^{2+\delta}) < \infty$  for some  $\delta > 0$ , that  $\sigma^2$  and  $p$  are continuous at  $x$  and that  $p(x) > 0$ . If  $h \rightarrow 0$  and  $nh \rightarrow \infty$  as  $n \rightarrow \infty$ , then

$$\text{var}(\hat{m}(x)|\mathbb{X}_n)^{-1/2} (\hat{m}(x) - \mathbb{E}(\hat{m}(x)|\mathbb{X}_n)) \rightsquigarrow N(\mathbf{0}, \mathbf{I}_{k+1}),$$

where  $\mathbf{I}_{k+1}$  denotes the  $(k+1) \times (k+1)$  identity matrix.

# Asymptotic distribution

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$$\begin{aligned} & \text{var}(\hat{\mathbf{m}}(x)|\mathbb{X}_n)^{-1/2} (\hat{\mathbf{m}}(x) - E(\hat{\mathbf{m}}(x)|\mathbb{X}_n)) \\ &= \left[ \frac{\sigma^2(x)p(x)}{nh} \cdot \mathbf{M}(x)^{-1} \mathbf{S}(x) \mathbf{M}(x)^{-1} + o_p\left(\frac{1}{nh}\right) \right]^{-1/2} \cdot [p(x)\mathbf{M}(x) + o_p(1)]^{-1} \cdot \frac{1}{n} D_x^\top W_x (Y - m) \\ &=: \mathbf{Z}_n(x)(1 + o_p(1)) \end{aligned}$$

For  $a \in \mathbb{R}^{k+1}$ , define  $\zeta_{ni}(x)$  by  $a^\top Z_n(x) = \sum_{i=1}^n \zeta_{ni}(x)$ ,

$$\begin{aligned} D_x^T W_x (Y - m) &= \left( \sum_{i=1}^n K_h(X_i - x) \frac{((X_i - x)^l}{l! k^l} (Y_i - m(X_i)) \right)_l \\ \zeta_{ni}(x) &= \left( \sum_{i=1}^n \sum_{l=0}^k b_l K_h(X_i - x) \frac{((X_i - x)^l}{l! k^l} (Y_i - m(X_i)) \right) \end{aligned}$$

# Asymptotic distribution

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$$E\zeta_{ni}(x) = 0$$

$$E\zeta_{ni}^2(x) = \frac{1}{h}(K^2)_h * \left( \sum_{l=0}^k \frac{b_l(\cdot - x)^l}{l!h^l} \right)^2 \sigma^2(x)p(x) = \frac{\sigma^2(x)p(x)}{h} (b^T \mathbf{S}(x)b + o_p(1))$$

$$\begin{aligned} b^T b &= a^T \left[ \mathbf{M}(x)^{-1} \mathbf{S}(x) \mathbf{M}(x)^{-1} \right]^{-1/2} \mathbf{M}(x)^{-1} \mathbf{M}(x)^{-1} \left[ \mathbf{M}(x)^{-1} \mathbf{S}(x) \mathbf{M}(x)^{-1} \right]^{-1/2} a \frac{h}{n\sigma^2(x)p(x)} \\ &= a^T a \cdot \text{tr}(\mathbf{S}^{-1}(x)) \frac{h}{n\sigma^2(x)p(x)} \end{aligned}$$

Therefore, for any  $a \in \mathbb{R}^{k+1}$ ,

$$s_n^2(x) = \sum_{i=1}^n \text{var}(\zeta_{ni}(x)) = \frac{n}{h} \text{tr}(\mathbf{S}(x)) b^T b = a^T a + o_p(1)$$

# Asymptotic distribution

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$$\begin{aligned}\sum_{i=1}^n E[|\zeta_{ni}(x)|^{2+\delta}] &= n \left( \sqrt{\frac{h}{n}} \right)^{2+\delta} E[(Y_1 - m(X_1))^{2+\delta} K_h(X_1 - x)^{2+\delta} \left( \sum_{l=0}^k \frac{b_l(X_1 - x)^l}{l! h^l} \right)] \\ &\sim (nh)^{-\delta/2} (K^{2+\delta})_h^- * \left( \sum \frac{b_l(\cdot - x)^l}{l! h^l} \right) p(\cdot) \\ &\sim (nh)^{-\delta/2} \rightarrow 0\end{aligned}$$

For any  $a \in \mathbb{R}^{k+1}$ ,

$$\begin{aligned}a^T Z_n(x) &\xrightarrow{d} N(0, a^T a) \\ Z_n(x) &\xrightarrow{d} N(0, I_{k+1})\end{aligned}$$

# Asymptotic distribution

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$$\begin{aligned}\hat{\mathbf{m}}(x) - \mathbf{m}(x) &= \hat{\mathbf{m}}(x) - E[\hat{\mathbf{m}}(x)|\mathbb{X}_n] + E[\hat{\mathbf{m}}(x)|\mathbb{X}_n] - \mathbf{m}(x) \\ &= \left( n^{-1}h^{-1} \cdot M(x)^{-1}S(x)M(x)^{-1} \cdot \frac{\sigma^2(x)}{\rho(x)} \right)^{1/2} \cdot V_n(x) \\ &\quad + h^{(k+1)} \cdot \frac{1}{(k+1)!} \cdot M(x)^{-1}\mu(x) \cdot m^{(k+1)}(x) \\ &\quad + o_p\left(h^{k+1} + n^{-1/2}h^{-1/2}\right),\end{aligned}$$

where  $V_n \xrightarrow{d} \mathcal{N}(0, I_{k+1})$ .